Year 12 Further Maths - Chi-squared tests

Similar questions - Contingency tables

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A school pupil keeps a note of whether her journeys to school and from school are delayed. The results for a random sample of journeys are shown in the table.

	To school	From school
Delayed	52	38
Not delayed	78	92

Test at the 10% significance level whether there is association between delays and the direction of the journey. **[7]**

2 For a random sample of 160 employees of a large company, the principal method of transport for getting to work, arranged according to the grade of the employee, is shown in the table.

Grade	Walk or cycle	Private motorised transport	Public transport
A	12	20	8
B	25	31	24
C	18	9	13

A test is carried out at the 5% significance level of whether there is association between grade of employee and method of transport.

(a) State appropriate hypotheses for the test. **[1]**

The contributions to the test statistic are shown in the following table, correct to 3 decimal places.

Grade	Walk or cycle	Private motorised transport	Public transport
A	0.223	1.667	0.939
B	0.227		0.100
C	1.314	2.400	0.272

(b) Show how the missing contribution (0.033) is obtained. **[3]**

(c) Complete the test, stating the conclusion. **[3]**

(d) Which combination of grade of employee and method of transport most strongly suggests association? Justify your answer. **[1]**

3 Fifty students from each of three schools P, Q and R took a mathematics test. The numbers who passed and failed are shown.

	School P	School Q	School R
Pass	34	41	25
Fail	16	9	25

(a) Test at the 1% significance level whether the pass rate is associated with the school. **[7]**

(b) State the number of degrees of freedom you used, and explain how it is found. **[1]**

(c) Identify the cell that contributes most to the test statistic and comment on what it shows. **[2]**

Answers

Question 1

H_0: no association between delays and direction; H_1: there is association. Row totals 90 and 170; column totals 130 and 130; total 260. Expected: delayed 45, 45; not delayed 85, 85. $\chi^2 = \frac{7^2}{45} + \frac{7^2}{45} + \frac{7^2}{85} + \frac{7^2}{85} = 1.089 + 1.089 + 0.576 + 0.576 = 3.33$. $\nu = (2-1)(2-1) = 1$; 10% critical value 2.706. $3.33 > 2.706$, so reject H_0: there is evidence at the 10% level of association between delays and direction

Question 2

(a) H_0: no association between grade and method of transport; H_1: there is association

(b) Row total $B = 80$, column total (private) $= 60$, total 160. $E = \frac{80 \times 60}{160} = 30$. Contribution $= \frac{(31-30)^2}{30} = 0.0333 \approx 0.033$

(c) $\chi^2 = 0.223 + 1.667 + 0.939 + 0.227 + 0.033 + 0.100 + 1.314 + 2.400 + 0.272 = 7.175$. $\nu = (3-1)(3-1) = 4$; 5% critical value 9.488. $7.175 < 9.488$, so do not reject H_0: there is insufficient evidence of association between grade and method

(d) Grade C with private motorised transport: it has the largest contribution (2.400); observed 9 is well below expected 15

Question 3

(a) H_0: pass rate is independent of school; H_1: associated. Row totals 100, 50; E: pass 33.33 each, fail 16.67 each. Contributions: 0.013, 1.763, 2.083, 0.027, 3.527, 4.167. $\chi^2 = 11.58$. $\nu = (2-1)(3-1) = 2$; 1% critical value 9.210. $11.58 > 9.210$, so reject H_0: there is evidence at the 1% level that pass rate is associated with school

(b) $\nu = (\text{rows} - 1)(\text{columns} - 1) = (2-1)(3-1) = 2$

(c) School R, fail: contribution 4.167 (observed 25 against expected 16.67): school R has more fails than would be expected if the pass rate were independent of school